*fast*Rise™

Multilayer Non-Reinforced Prepreg

AGC
Your Dreams, Our Challenge

Benefits

- Df = 0.0014 / 0.0017 (10/40 GHz)
- Laser ablatable, HDI ready
- Low Dk enables reduced thickness of ATE boards
- Low temperature alternative to thermoplastic films in military designs
- Multilayer prepreg for high layer count high speed digital
- Stable Dk over temperature
- Fiberglass free prepreg
- Enables 5+ sequential laminations
- Compatible with conductive pastes between subassemblies



Applications

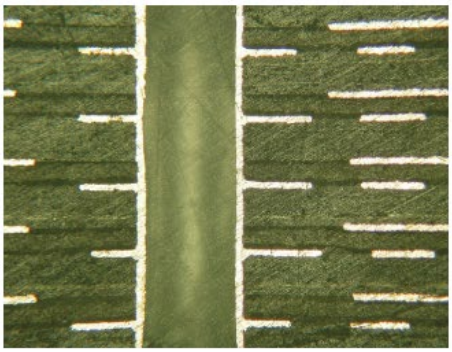
- Filters, Couplers
- Avionics & aerospace
- Automotive Radar
- Beam Steerable Antennas
- Flexible Circuits

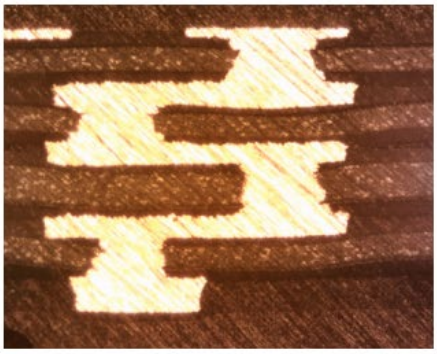
*fast*Rise™ is designed to bond all manner of circuit boards together with the lowest possible loss of any thermosetting prepreg available today. *fast*Rise™ enables 77 GHz automotive radar. *fast*Rise™ is non-reinforced and eliminates skew/variation in high speed digital/RF circuits. *fast*Rise™ is based on ceramic, thermoset and PTFE and is ideal for use with AGC's TSM-DS3, TSM-DS3b, TSM-DS3M and EZ-IO-F. *fast*Rise™ can be foil laminated, laser ablated, and sequentially laminated to yield layers of stacked or staggered microvias. Please consult with a technical sales person for microvia design and reliability testing data. The low temperature lamination leads to less variation when resistive foils are used. *fast*Rise™ is flexible enough in the prepreg stage for lasing and impregnation with conductive paste for interconnecting between subassemblies. When combined with stable dielectric materials like EZ-IO-F, *fast*Rise™ passes reliability testing including IST, HATS and CAF.

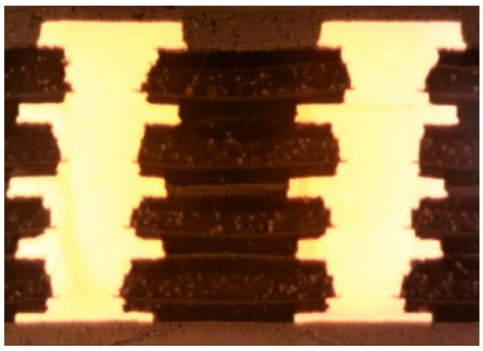
The low 420 °F (215 °C) lamination temperature enables 5+ sequential laminations at temperatures lower than those normally used for FEP and PFA in Avionics & aerospace constructions. The *fast*Rise™ bonds to PTFE, epoxy, low-flow epoxy, LCP, polyimide, and hydrocarbon materials.
There are many *fast*Rise™ part numbers due to the diverse number of tasks that a prepreg must fulfill. For doing sequential laminations where the intent is to flow and fill copper that has been plated up to 2-3 mils, high resin content and high flow is required from the prepreg. For applications where you do not want the prepreg flowing into cavities, a low flow prepreg is desirable.

Extremely high flow is needed for filling blind or buried vias or milled out cavities. In some coupler designs, a very thin prepreg is desirable for maximum coupling between the overlay couplers and sufficient flow is only needed to bond artwork with 0.5 oz. copper. Low flow prepregs are best for doing foil laminations. Flow and fill requirements don't exist in a foil lamination so a low flow prepreg is appropriate whereas a high flow prepreg might be more prone to cosmetic defects. AGC has found that low flow prepregs are most suitable for microvia formation following a foil lamination (the microvia cross sections below are courtesy of Hughes Circuits).

AGC's low flow prepregs have a much better lased hole quality than the high flow prepregs. In a high layer count PWB, often there are many layers of overlapping edge-coupled traces. High layer count PWBs are susceptible to lamination voids due to areas of high and low pressure. For all of these reasons it is best to consult the *fast*Rise™ design guide or talk to an AGC applications engineer to design with the most suitable prepreg.

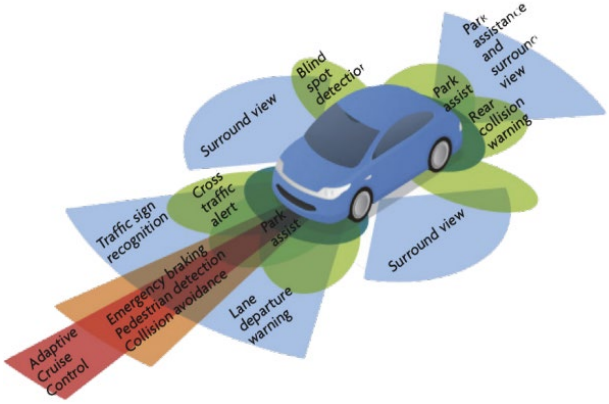






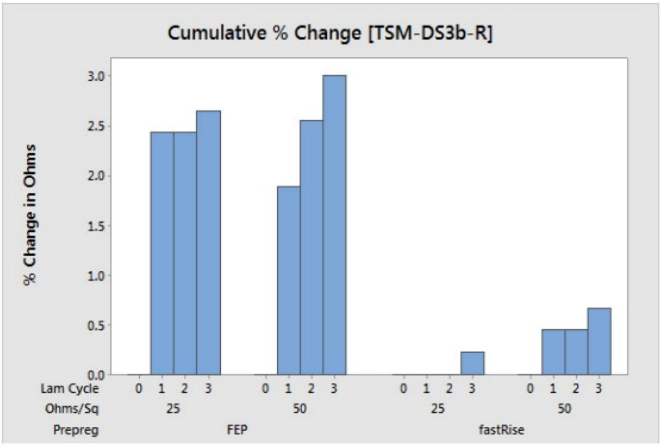


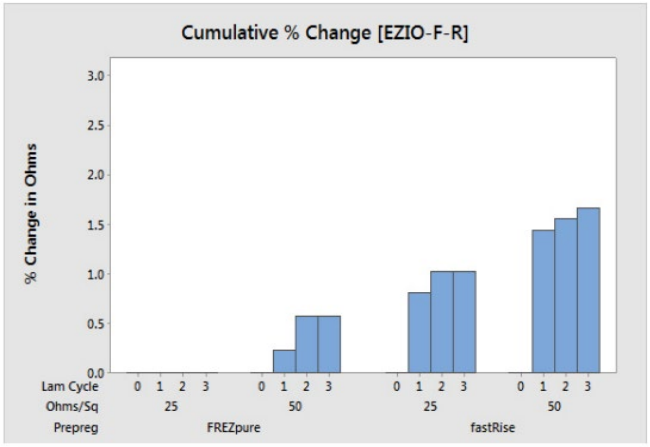
*fast*RiseTM prepreg between subassemblies w/copper paste



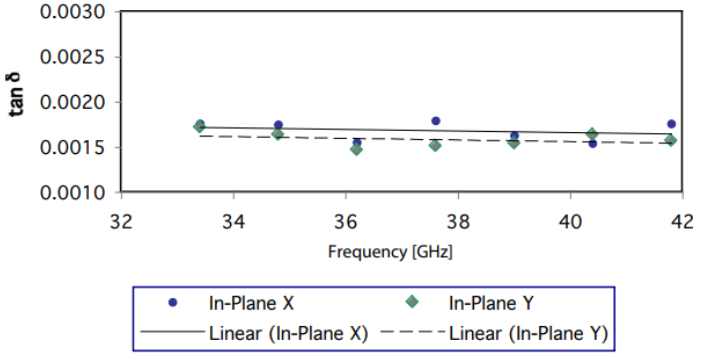
*fast*Rise™ used at 77 GHz for Automotive Radar

EZ-IO-F-R / TSM-DS3b-R Resistor Foil Stability with Prepreg Lamination

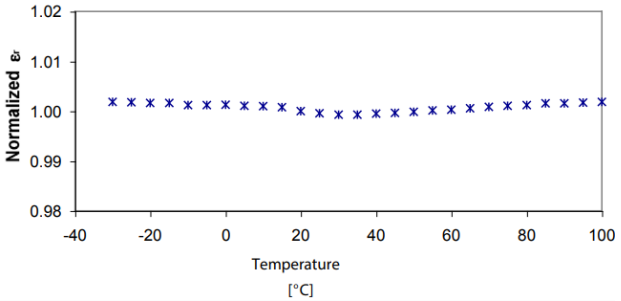




*fast*Rise™ Dielectric Loss at mmWave Frequency



Temperature Influence on Dielectric Constant



*fast*Rise™ is shipped at a very low degree of cure. The best flow conditions are achieved when the prepreg spends the maximum amount of time possible at a temperature of 225 °F (107 °C) using the highest possible pressure. For difficult flow and fill designs or when using AGC's low flow prepregs, the lamination press should ramp up to 225 °F (107 °C) and hold for 30-60 minutes at maximum pressure, followed by a slow 2.0-4.0 °C/min. ramp rate to 420 °F (216 °C).
Because of the large number of possible applications for the *fast*Rise™ prepreg series and the complexity of many multilayer printed circuit designs, AGC does not warranty or guarantee the performance of *fast*Rise™ when combined with any supplier's core materials. It is the responsibility of the end user to determine the suitability of *fast*Rise™ for any application.

AGC's *fast*Rise™ materials meet the requirements of IPC-4103B/520 and IPC-4103B/530

Properties	Conditions	Typical Value	Unit	Test Method
Electrical Properties				
Dielectric Constant	@ 10 GHz	2.43 – 2.77		IPC-650 2.5.5.5.1 (modified)
Dissipation Factor	@ 10 GHz	0.0010 – 0.0015		IPC-650 2.5.5.5.1 (modified)
Volume Resistivity		8.00×10^8	Mohms/cm	IPC-650 2.5.17.1 (after elevated temp)
		1.71×10^8	Mohms/cm	IPC-650 2.5.17.1 (after humidity)
Surface Resistivity		3.48×10^8	Mohms	IPC-650 2.5.17.1 (after elevated temp)
		1.16×10^8	Mohms	IPC-650 2.5.17.1 (after humidity)
Thermal Properties				
Thermal Conductivity		0.25	W/M*K	ASTM F433
CTE (-55 to125°C)	X	59	ppm/°C	IPC-650 2.4.41/TMA
	Y	70	ppm/°C	
	Z	72	ppm/°C	
T_d	2% wt. loss	376 (709)	°C (°F)	IPC-650 2.4.24.6/TGA
	5% wt. loss	421 (790)	°C (°F)	
TcK (-30 to 100 °C)		0.06	ppm/°C	IPC-650 2.5.5.5.1 (modified)
T_g		188	°C	ASTM E 1640 (DMA)
Mechanical Properties				
Peel Strength	HH/H1	0.35-0.53 (2-3)	N/mm (lbs/in)	IPC-650 2.4.8
Dielectric Strength		42.9 (1,090)	Kv/mm (V/mil)	ASTM D 149
Tensile Strength	x	12 (1,690)	N/mm^2 (psi)	ASTM D 882
	y	10 (1,480)	N/mm^2 (psi)	
Tensile Modulus	x	2,100 (304)	N/mm^2 (kpsi)	ASTM D 882
	y	2,030 (295)	N/mm^2 (kpsi)	
Elongation at Break	x	0.82	%	ASTM D 882
	y	0.73	%	
Chemical / Physical Properties				
Moisture Absorption		0.08	%	IPC-650 2.6.2.1
Dielectric Breakdown		49	Kv	IPC-650 2.5.6 (parallel to lamination)
Density	Specific Gravity	1.82	g/cm^3	ASTM D 792 Method A
Hardness		68	Shore D	ASTM D 2240

Standard *fast*Rise™ Part Numbers							
Product	Stripline with 1 oz. Cu	High Layer Count PWBs	Between Plated Up Subassemblies	Fill Blind/ Buried Vias	Resin Content	Microvia Formation/Foil Lamination	Drill Quality
FR-27-0040-43F	See below [2]	Yes[1]	See below [3]	No	Medium	Best	Best
FR-27-0045-35	See below [2]	Yes[1]	No	No	High	R	Best
FR-28-0040-50	Yes	Yes	See below [3]	No	High	R	Best
FR-27-0050-40	Yes	Yes	See below [3]	No	High	R	Best

Specialty *fast*Rise™ Part Numbers							
Product	**Stripline with 1 oz. Cu**	**High Layer Count PWBs**	**Between Plated Up Subassemblies**	**Fill Blind/ Buried Vias**	**Resin Content**	**Microvia Formation/Foil Lamination**	**Drill Quality**
FR-25-0021-45	No	No[1]	No	No	Low	No	Susceptible[4]
FR-25-0021-45F	No	No[1]	No	No	Low	No	Susceptible[4]
FR-26-0025-60	Yes	Yes[1]	See below[3]	No	High	No	Susceptible[4]
FR-27-0035-66	Yes	Yes[1]	Yes	Yes	High	No	Susceptible[4]
FR-27-0042-75	Yes	Yes[1]	Yes	Yes	Highest	No	Susceptible[4]

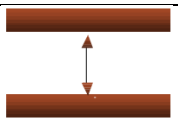
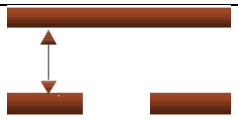
[1] Some layers only

[2] FR-27-0030-25, FR-27-0040-43F and FR-27-0045-35 can be used with 1 oz. copper on low layer count PWBs but should not be used where many layers are bonded together in a single lamination due to the risk of low pressure areas during lamination.

[3] Plated up subassemblies can vary in the ultimate copper thickness. A discussion with an AGC Applications Engineer is advised.

[4] Susceptible to common PTFE drilling defects

R = Recommended

*fast*Rise™ Prepreg					
Product	**Nominal Dk (10 GHz)**	**Pressed Thickness¤ (mil)**	**Pressed Thickness¤ (mil) 0.5 oz. Cu**	**Pressed Thickness¤ (mil) 1 oz. Cu**	**Typical Flow %**
FR-25-0021-45	2.43	2.1	1.9	1.5	10.0
FR-25-0021-45F	2.45	2.1	1.9	1.5	2.0
FR-26-0025-60	2.52	2.8	2.5	2.1	11.0
FR-27-0030-25	2.70	3.6	3.2	2.8	3.0
FR-27-0035-66	2.62	3.9	3.6	3.2	15.0
FR-27-0040-43F	2.77	4.1	3.8	3.5	3.0
FR-27-0042-75	2.68	5.0	4.7	4.3	30.0
FR-27-0045-35	2.72	5.6	5.2	4.8	7.0
FR-27-0050-40	2.74	6.0	5.6	5.2	11.0
FR-28-0040-50	2.74	4.4	4.1	3.7	10.0
			 0.5 oz. Cu, 50% removal	 1 oz. Cu, 50% removal	

¤ : Pressed between ground planes * Insufficient data

* All test data provided are typical values and not intended to be specification values. For review of critical specification tolerances, please contact a company representative directly.
* Please contact AGC for availability of additional thicknesses, other sizes.

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